

Assignment 6

1. Multiple Choice Questions (Only one correct answer per question)

- Which is a state function?
 - Heat (q)
 - Work (w)
 - ☒ Internal energy (E)
 - Path length
- In the first law $\Delta E = q + w$, w is positive when:
 - System expands
 - ☒ Surroundings do work on system
 - System does work on surroundings
 - Volume is constant
- Work in L·atm for expansion from 2.00 L to 5.00 L at 1.00 atm:
 - ☒ +3.00
 - ☐ -3.00
 - +5.00
 - 5.00
- At constant pressure, $q_p =$:
 - ΔE
 - ☒ ΔH
 - $-\Delta E$
 - $-\Delta H$

5. An endothermic process:
- a) Involves negative ΔE
 - b) Releases heat
 - ☒ c) Absorbs heat
 - d) Involves negative ΔH
6. Calorimetry in a coffee-cup measures:
- a) ΔE at constant volume
 - ☒ b) ΔH at constant pressure
 - c) ΔE at constant pressure
 - d) ΔH at constant volume
7. The bomb calorimeter measures:
- a) Heat at constant P
 - ☒ b) Heat at constant V
 - c) Latent heat only
 - d) Work directly
8. In Hess's Law, ΔH is:
- ☒ a) Path dependent
 - ☒ b) Path independent
 - c) Proportional to path length
 - d) Undefined
9. ΔH°_f of elements in their standard state is:
- ☒ a) 0 kJ/mol
 - b) 1 kJ/mol
 - c) 100 kJ/mol
 - d) Varies
10. $\Delta H^\circ_{\text{rxn}}$ from ΔH°_f data:
- ☒ a) $\sum \Delta H^\circ_f(\text{products}) - \sum \Delta H^\circ_f(\text{reactants})$
 - b) $\sum \Delta H^\circ_f(\text{reactants}) - \sum \Delta H^\circ_f(\text{products})$
 - c) Average of reactants and products
 - d) None

11. Conversion $1 \text{ L} \cdot \text{atm} = ? \text{ J}$

- ☒ a) 101.3
- b) 4.184
- c) 8.314
- d) 273

12. In $\Delta E = q + w$, for adiabatic expansion of an ideal gas:

- a) $q > 0$
- ☒ b) $q = 0$
- c) $w = 0$
- d) $\Delta E > 0$

13. Increasing reactant amounts by factor n changes ΔH_{rxn} by:

- a) $\Delta H/n$
- ☒ b) $n\Delta H$
- c) No change
- d) Cannot predict

14. Which statement about spontaneous processes is correct?

- a) Always fast
- b) Always exothermic
- ☒ c) Occur without continuous external input once started
- d) Have $\Delta S = 0$

15. For a reversible process:

- a) $\Delta S_{\text{univ}} > 0$
- ☒ b) $\Delta S_{\text{univ}} = 0$
- c) $\Delta S_{\text{univ}} < 0$
- d) $\Delta S_{\text{sys}} = 0$ always

16. An increase in entropy is observed for:

- a) Freezing water
- b) Condensation

- ☒ c) Vaporization
- d) Crystallization

17. The second law of thermodynamics states:

- a) $\Delta E_{\text{univ}} > 0$
- ☒ b) $\Delta S_{\text{univ}} > 0$ for spontaneous processes
- c) $\Delta H_{\text{univ}} > 0$
- d) $\Delta G_{\text{univ}} > 0$

18. Which has the greatest standard molar entropy at 298 K?

- a) $\text{H}_2\text{O}(\text{s})$
- b) $\text{H}_2\text{O}(\text{l})$
- ☒ c) $\text{H}_2\text{O}(\text{g})$
- d) All the same

19. Standard molar entropy values are:

- a) Zero for all substances at 0 °C
- ☒ b) Always positive above 0 K
- c) Negative for gases
- d) Larger for elements than compounds

20. For the reaction: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$, ΔS° is:

- a) Positive
- ☒ b) Negative
- c) Zero
- d) Cannot tell

21. $\Delta G < 0$ means:

- ☒ a) Spontaneous at given conditions
- b) Nonspontaneous
- c) At equilibrium
- d) Endothermic

22. If $\Delta H < 0$, $\Delta S > 0$, reaction is:

- ☒ a) Spontaneous at all T
- b) Nonspontaneous at all T
- c) Spontaneous only at high T
- d) Spontaneous only at low T

23. If $K = 1$, $\Delta G^\circ =$:

- a) Positive
- ☒ b) Zero
- c) Negative
- d) Undefined

24. If $\Delta G^\circ = -10 \text{ kJ/mol}$ at 298 K, K is:

- ☒ a) >1
- b) <1
- c) $=1$
- d) Cannot tell

25. Entropy change for surroundings at constant P:

- ☒ a) $\Delta S_{\text{surr}} = -\Delta H_{\text{sys}} / T$
- b) $\Delta S_{\text{surr}} = \Delta H_{\text{sys}} / T$
- c) $\Delta S_{\text{surr}} = \Delta H_{\text{sys}} / P$
- d) $\Delta S_{\text{surr}} = \Delta E_{\text{sys}} / T$

26. Third law says:

- ☒ a) S of perfect crystal = 0 at 0 K
- b) S of perfect crystal = finite at 0 K
- c) $\Delta S = 0$ for all processes
- d) S is independent of T

27. Highest ozone concentration is in the:

- a) Troposphere
- ☒ b) Stratosphere
- c) Mesosphere
- d) Thermosphere

28. Ozone forms when:

- ☒ a) $\text{O}_2 \rightarrow 2\text{O}$ by UV light, O combines with O_2
- b) O_2 absorbs IR
- c) O_3 absorbs UV-B
- d) N_2 dissociates

29. CFCs deplete ozone because they:

- a) Absorb UV directly
- ☒ b) Release $\text{Cl}\cdot$ radicals upon photolysis
- c) Increase CO_2
- d) Produce SO_2

30. Acid rain from SO_2 involves intermediate formation of:

- a) H_2SO_4 directly
- b) H_2O_2
- ☒ c) SO_3
- d) H_2S

31. Greenhouse gas with highest present-day atmospheric concentration is:

- ☒ a) CO_2
- b) CH_4
- c) N_2O
- d) CFC-12

32. Eutrophication in lakes is caused mainly by:

- a) Oil spills
- ☒ b) Nutrient runoff (nitrates, phosphates)
- c) Thermal pollution
- d) PCB contamination

33. Increased CO_2 affects ocean chemistry by:

- a) Increasing pH

- b) Decreasing pH (acidification)
- c) Creating alkaline water
- d) Removing bicarbonates

34. Which of the following is not the 12 principles associated with green chemistry?

- a) chemical products should be designed to minimize toxicity **even at the cost of their desired function** 4
- b) the end products of chemical processing should break down at the 10 end of their useful lives into innocuous degradation products
- c) promotion of the use of catalysts 9
- d) inherently safer chemistry for accident prevention

35. The visible reddish-brown component of photochemical smog is:

- a) NO₂
- b) O₃
- c) PAN
- d) CH₄

36. Which step in water treatment kills pathogens?

- a) Sedimentation
- b) Filtration
- c) Chlorination or ozonation
- d) Aeration

2. Short-Answer/Calculation Questions

1. A steel cylinder absorbs 15.5 kJ heat and does 4.2 kJ work. Find ΔE .

$$\Delta E = q + w = (15.5 \text{ kJ}) + (-4.2 \text{ kJ}) = 11.3 \text{ kJ}$$

2. For reaction with ΔH°_f data: H₂O(l): -285.8 kJ/mol, H₂(g): 0, O₂(g): 0.

Find ΔH° for $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O(l)}$. $\Delta H^\circ_f = H_{\text{product}} - H_{\text{reactant}} = 2(-285.8) - 2(0) - 0 = -571.6 \text{ kJ/mol}$

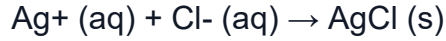
3. Calculate $\Delta S^\circ_{\text{rxn}}$ at 298 K for $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$, given

$S^\circ(\text{N}_2) = 191.5$, $S^\circ(\text{H}_2) = 130.6$, $S^\circ(\text{NH}_3) = 192.8 \text{ J/mol}\cdot\text{K}$. $\Delta S^\circ = \sum S^\circ_{\text{products}} - \sum S^\circ_{\text{reactants}} = 2(192.8) - (191.5 + 3 \cdot 130.6) = -197.7 \text{ J/mol}\cdot\text{K}$

4. Determine spontaneity: $\Delta H = -50 \text{ kJ}$, $\Delta S = -100 \text{ J/K}$ at 298 K.

5. Consider the reaction:

$$\Delta G = \Delta H - T\Delta S = -50 \times 10^3 - 298(-100) = -20200 \text{ J} \rightarrow \text{Spontaneous in forward direction}$$



Given the following table of thermodynamic data at 298 K:

Substance	ΔH_f° (kJ/mol)	S° (J/K·mol)
$\text{Ag}^+ (\text{aq})$	105.90	73.93
$\text{Cl}^- (\text{aq})$	-167.2	56.5
$\text{AgCl} (\text{s})$	-127.0	96.11

$$\therefore \Delta G^\circ = \Delta H^\circ - T\Delta S^\circ = -65.7 \times 10^3 - 298 \cdot (-34.32) \text{ J/mol}$$

$$= -55472.64 \text{ J/mol}$$

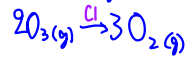
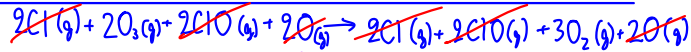
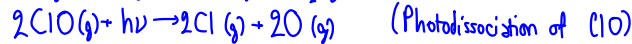
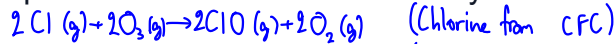
$$K = e^{-\Delta G^\circ / RT} = e^{-\frac{-55472.64}{8.31 \cdot 298}} = 5.35 \times 10^9$$

Find the value of K for the reaction at 25 °C.

$$\Delta H_{\text{rxn}}^\circ = -127.0 - (-167.2 + 105.90) = -65.7 \text{ kJ/mol}$$

$$\Delta S^\circ = 96.11 - (56.5 + 73.93) = -34.32 \text{ J/mol}\cdot\text{K}$$

6. Explain how $\text{Cl}\cdot$ from CFCs destroys ozone catalytically.



→ Cl activates the conversion of O_3 into O_2 , which destroys and decrease the amount of O_3